Regi Kusumaatmadja

Düsseldorf Institute for Competition Economics (DICE)
Heinrich Heine University Düsseldorf
Email: kusumaatmadja@dice.hhu.de
Website: https://regikusumaatmadja.com

Employment

2026.10 - present, Postdoctoral Researcher, Düsseldorf Institute for Competition Economics (DICE), Heinrich Heine University Düsseldorf.

Education

VU Amsterdam and Tinbergen Institute, Ph.D. in Economics, Expected Completion: 2026.
Tinbergen Institute, M.Phil. in Economics, 2021.
VU Amsterdam, M.Sc. in Economics, 2018.
University of Indonesia, B.Sc. in Physics, 2016 (Best Graduate).

Letter Writers

prof. dr. José Luis Moraga-González
VU Amsterdam & Télécom Paris
Tinbergen Institute, CEPR, CESifo
Email: j.l.moragagonzalez@vu.nl

prof. dr. Sabien Dobbelaere
VU Amsterdam
Tinbergen Institute, IZA
Email: sabien.dobbelaere@vu.nl

prof. dr. Mark J. Roberts
Pennsylvania State University
NBER, ZEW
Email: mroberts@psu.edu

prof. dr. Eric J. Bartelsman (placement dir.)
VU Amsterdam
Tinbergen Institute, IZA
Email: e.j.bartelsman@vu.nl

Visiting Position

2026.05 - 2026.06, CUNEF Universidad Madrid, invited by **dr. Costanza Cincotta**
2023.01 - 2023.06, the Pennsylvania State University, invited by **prof. dr. Mark J. Roberts**

Research and Teaching Areas

Primary Fields: Industrial Organization, Economics of Innovation
Secondary fields: Applied Econometrics, Applied Microeconomics

Research in Progress

Internal and External R&D: An analysis of costs and benefits

This paper analyzes the costs and benefits of in-house and contracted-out R&D. Estimating a dynamic model on Dutch ICT firms from 2000 to 2020, I find that productivity gains are similar across R&D activities, but that contracting R&D out is much more costly than doing it in-house. This explains why so few firms only contract out. However, for a firm that also does in-house R&D, 94% of this extra cost disappears, which I interpret as a hold-up problem in the external contract. Finally, a uniform subsidy raises R&D participation and firm value more than a subsidy for in-house R&D only.

Presentation: JEI - Gran Canaria (2022), Ph.D.-EVS (2022), IO/Applied Micro Brownbag - Penn State (2023), VU Amsterdam Internal Seminar (2023), EARIE - Rome (2023), Tinbergen Institute Ph.D. Seminar (2023), DKIIS - Valencia (2023), CAED - Penn State (2024), SED Meeting - Barcelona (2024), CEPR Workshop: Trade, Geography, and Industrial Organization - Erasmus Rotterdam (2024, poster), APIOC - Seoul (2024), CEPR Symposium - Paris (2024, poster), EEA-ESEM - Bordeaux (2025), CEPR IO Virtual Gathering (2025)

Entry into Innovation Areas

This paper investigates the entry decisions of U.S. electronics firms across innovation areas. I apply topic modeling to over 630,000 patents to endogenously define innovation areas and merge those areas with market-implied patent valuations. Firms occupy several areas at once, and the number of firms differs widely across areas. An entry game, estimated by moment inequalities, finds that additional rivals reduce expected profit and that firms gain from occupying more areas at once. In the case study of semiconductor, a payment based on a quarter of predicted value shifts 0.1–0.5% into those areas and leads up to 0.8% of firms to add an extra area. Meanwhile, a 25% reduction in the occupancy cost shifts up to 7% into those areas and increases up to 10% of firms to add an extra area.

Presentation: VU Amsterdam Internal Seminar (2025)

The Impact of Government Financial Assistance on the Scope and Direction of Firm Innovation, with C. Cincotta.

Governments spend hundreds of billions of dollars each year financing private innovation. We ask whether this spending expands, redirects, or reallocates research. We take a Salop model to U.S. public firms from 2000 to 2021. We compare recipients with close and distant rivals using Good Jobs First awards, patent data, and text-based product-market overlap. Distant rivals provide the preferred reading of the treated firm's own response. Recipients' patenting scale rises by about 4 percent relative to distant product-market controls. Nearby rivals do not change how much they patent. Direction shifts toward the industry's most-cited technologies, strongest for cash grants.

Mergers and Innovation: An Exploration of Scope and Direction of Innovation, with S. Dobbelaere and J. L. Moraga-Gonzalez.

How do mergers change the scale and composition of firm innovation? We study U.S. public–public mergers from 1980 to 2020 using combined acquirer–target patent portfolios. A compact framework isolates business stealing, knowledge pooling, and activation, and keeps product-market overlap distinct from technological overlap. We compare treated merger pairs to matched control pairs before and after the deal. On the matched sample, joint-portfolio scope is larger after mergers, by about 0.18 on the annual measure. The difference is larger when pre-merger technologies overlap, especially among non-horizontal deals. Joint portfolios also show higher alignment with highly cited patents.

Presentation: VU Amsterdam Internal Seminar (2024), Tinbergen Institute Workshop in Economics of Innovation (2025)

Teaching Experience

2021-2025 TA, VU Amsterdam, Introduction to Econometrics (MSc). Eval: 4.3/5.0
2021-2025 TA, VU Amsterdam, Applied Econometrics (MSc). Eval: 4.7/5.0
2022-2025 Supervisor, VU Amsterdam, Research Project and Thesis (BSc + MSc).
2021-2024 Instructor, VU Amsterdam, Math, Statistics, Micro, and R (Refresher MSc).
2022 TA, VU Amsterdam, Microeconomics I (BSc).
2021-2025 Guest Lecturer, VU Amsterdam, Urban Economic Challenges & Policies (MSc).

Relevant Experience

2023, EARIE 1st Summer School on Entry Model, LUISS - Rome.
2022, Econometric Society's Dynamic Structural Econometrics Summer School, MIT.
2019, Research on Productivity, Trade, and Growth Summer School, Tinbergen Institute.
2020-2021, RA for Eric J. Bartelsman and Sabien Dobbelaere, VU Amsterdam, MICROPROD project.
2018-2019, Research Associate, Institute for Economic & Social Research, University of Indonesia (LPEM FEB UI).

Scholarships, Honors, and Awards

2025, Excellence in Teaching Award, VU Amsterdam.
2023, Research Visit Grant, VU Amsterdam (monetary grant).
2019-2021, Tinbergen Institute - Full Scholarship (tuition & living cost).
2017, VU Amsterdam Fellowship Programme (tuition).
2017, Holland Scholarship Programme (living cost).
2016, Best Graduate, Department of Physics, University of Indonesia (monetary prize).
2016, Top 1 (one) Percent, Faculty of Mathematics and Natural Sciences, University of Indonesia.
2014-2016, XL-Axiata Future Leaders Scholarship (leadership programme & in-kind benefits).
2014-2015, Karya Salemba Empat Scholarship (tuition & living cost).

Publications

Evaluating the congestion-reducing effects of road rationing policy: Evidence from Jakarta's odd-even policy. *Case Studies on Transport Policy* (2025), with M. Halley Yudhistira, Y. Sofiyandi, and M. Firman Hidayat. (previously circulated as Does Traffic Management Matter? Evaluating Congestion Effect of Odd-Even Policy in Jakarta)

Theoretical exploration of optical response of Fe_3O_4-reduced graphene oxide nano-particle system within dynamical mean-field theory. *IOP Conf. Ser.: Mater. Sci. Eng.* (2017), with M.A. Majidi, A.D. Fauzi, W.Y. Phan, A. Taufik, R. Saleh, and A. Rusydi.

Policy Work

The sky in blues: on the recent development of Indonesian airlines industry. *LPEM FEB UI WP* (2019), with C. Nuryakin, N. W. G. Massiw, A. U. Lumbanraja, S. Hambali, and D. Anindita

Miscellaneous

Language: Indonesian (native), English (fluent), Dutch (basic-intermediate), Arabic (basic-intermediate)
Software: R, Python, Rust

Last updated: October 26, 2026
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